What scaling up RHD screening and prophylaxis buys in Indonesia and Uganda, on provisional parameters

Technical review of the rheumatic heart disease investment-case model

William Garcia

University of Washington · World Bank

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We ask whether scaling up echo screening and prophylaxis is worth its cost

1. **Health:** how many rheumatic heart disease (RHD) deaths does scale-up avert in Indonesia and Uganda, 2026–2100?
2. **Services and cost:** what screening, prophylaxis and surgery does it add, at what cost?
3. **Value:** does the monetised benefit exceed the incremental cost?

Scope: national, both sexes, ages 0–100+; secondary prevention only.

About 2.2 million people live with RHD in the two countries, and few receive prophylaxis

2.16 million

people living with RHD in Indonesia and Uganda, 2023

Baseline optimal-treatment coverage in the model: 4% in Indonesia, 0.75% in Uganda.

GBD 2023 RHD prevalence, published counts summed over age and sex (Indonesia 1,400,743; Uganda 762,663). Amber marks provisional assumptions throughout.

The model extends an established RHD investment case, but its stage parameters are provisional

Builds on

- Investment-case design of Coates et al. (2021): in the African Union, secondary prevention with secondary and tertiary care returned a benefit–cost ratio (BCR) of 4.7 to 2030
- World Heart Federation (WHF) echo stages A–D (Rwebembera et al., 2023)
- GBD 2023 burden; UN World Population Prospects (WPP) 2024

Boundary

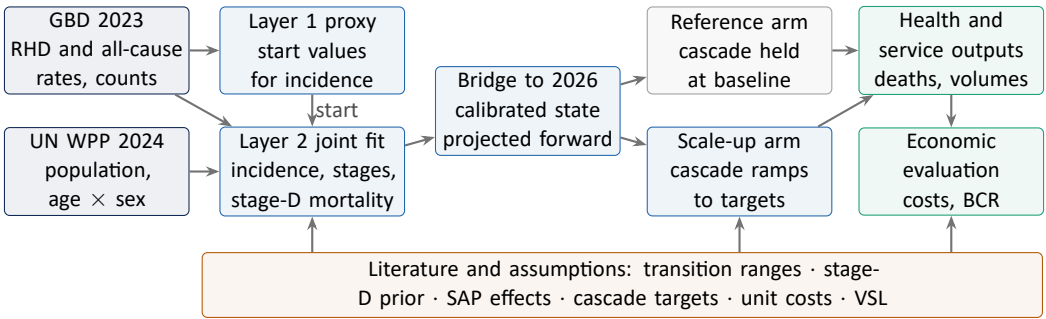
Deterministic point estimates; no sensitivity analysis yet.

Stage-D-only mortality and a progression effect of SAP are provisional structural choices.

Stage parameters are fitted to totals and only partly identified.

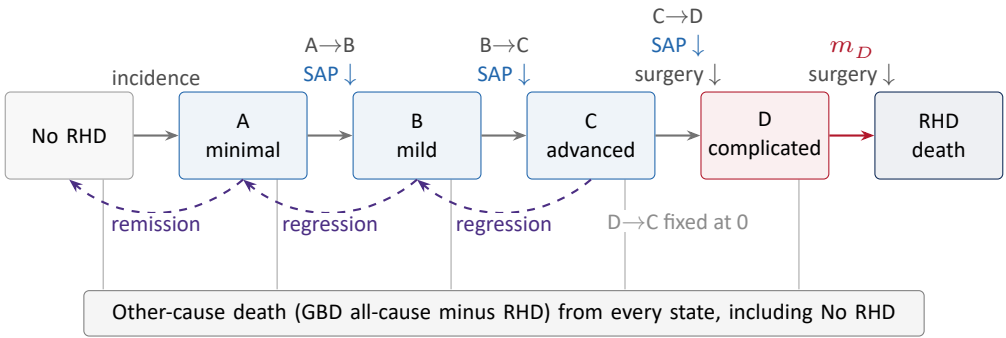
How the model works

Four linked steps turn GBD and WPP inputs into health and economic contrasts



GBD supplies the targets of both layers; both use WPP population as the denominator. Layer 2, the bridge and both arms run the same annual engine. SAP: secondary antibiotic prophylaxis. VSL: value of a statistical life.

Incident RHD enters stage A; SAP slows progression and only stage D carries RHD deaths



Drawn from the saved specification (D_only mortality; SAP mechanism progression). Surgery is a service acting on C→D and stage-D death, not a state.

Model stages follow WHF 2023, except that stage A spans all ages and only stage D can die of RHD

Stage	WHF 2023 definition	Model treatment
A	Minimal echo criteria; ages ≤ 20 only; low risk	Entry for all incident RHD, any age
B	Mild RHD; moderate or high risk	Exchanges with A and C
C	Advanced, at risk of complications	Progresses to D; regresses to B
D	Advanced with complications	Leaves only by death (D $\rightarrow$ C = 0)
RHD death	Not a stage	Only from D; A–C exactly zero

Definitions from Rwebembera et al. (2023); complications include heart failure, arrhythmia, stroke or need for surgery.

Population is exogenous: WPP fixes each age–sex cell, and RHD stocks age within it

Held fixed outside the model

- Population P_t: WPP 2024, ages 0–100+, two sexes
- Background deaths: GBD all-cause minus RHD (held at 2023 after calibration)
- Incidence: Layer-2 2019 rate per susceptible $\times$ trend $^{t-2019}$

Evolved by the model

- Stocks A–D by age and sex
- Susceptible pool $= P_t - (A + B + C + D)$
- Annual cycles, 2026–2100
- 2026 stock: calibrated 2019 state, bridged forward

Incidence trend: 0.985 per year (Indonesia), 0.985 (Uganda). The 2026 stock is 1,093,222 (Indonesia) and 784,769 (Uganda) people with RHD.

Competing exits are split on the hazard scale, so every stock balances exactly

Each stage faces several annual risks at once; converting them to hazards keeps every outflow non-negative.

$$p_{s\to e} = \left(1 - e^{-H_s}\right) \frac{h_e}{H_s}, \qquad h_e = -\ln\left(1 - p_e\right), \qquad H_s = \sum_{e \in \mathcal{E}_s} h_e$$

Stage s keeps the share e^{-H_s}. $\mathcal{E}_s$ holds the exits available from s — for stage A: to no RHD, to B, other-cause death (its RHD-death hazard is exactly zero). All probabilities are per year.

Each year draws incidence from the susceptible pool, applies exits, then ages survivors

$$A_{t+1}(a{+}1) = A_t(a)\, e^{-H_A} + B_t(a)\, p_{B\to A} + W_t(a)\, p_{W\to A}$$

$A_t(a)$: stage-A persons aged a entering year t, for each sex.
$W_t = P_t - (A_t + B_t + C_t + D_t)$: the susceptible pool, where incidence competes with other-cause death to give $p_{W\to A}$. Stages B–D follow the same form.

Observed data anchor demography and burden; stage dynamics are calibrated around provisional priors

Input	Source	Role	Status
Population	UN WPP 2024	Age × sex × year	Observed
Prevalence, deaths	GBD rate × WPP	Layer-2 targets	Observed
Background deaths	GBD all-cause — RHD	Competing risk	Observed
RHD incidence	GBD × Layer-2 α	Inflow to stage A	Calibrated
A–D transitions	Layer 2, literature bounds	Stage dynamics	Partly identified
Stage-D mortality	Layer 2, workbook prior	RHD deaths	Provisional
Incidence trend	Assumption	Scales incidence	Provisional

Effects, costs and valuation rest on literature and provisional values

Input	Value	Role	Status
SAP effect	HR 0.04 / 0.20 / 0.45	A→B, B→C, C→D	Provisional
Adherence	1 in both arms	Scales protection	No attenuation
Surgery effect	85% on two exits	Scaled by reach	Literature
Cascade targets	80% / 80% / 65%	Scale-up by 2050	Policy scenario
Screening schedule	Screens: 2 in 5–19, 1 in 20–39	Encounters per year	Assumption
Unit costs	Screening, SAP, surgery	Cost per service	Undocumented
GNI per capita	World Bank, 2019	Anchors VSL	Observed
VSL, discounting	30× GNI pc; 3%	Benefits and costs	Multiple provisional

Targets: screening reach / diagnosis / SAP coverage. Screening ages: WHO 2024 supports 5–19; the adult window and frequencies are assumptions. Adherence 1 = no further attenuation.

Calibration fits incidence, transitions and stage-D mortality jointly inside the production engine

Layer 1: aggregate proxy (2000–2019)

- Well–sick–dead, no remission; GBD counts
- 10 multipliers per sex: incidence, mortality
- Inverse-variance weights on
- Role: start values only

Layer 2: joint, production engine (scored 2010–2019)

- Prevalence : deaths weight 2.5 : 1; deaths all ages
- Free: 19 parameters (10 incidence α, 6 transitions, 3 other)
- Zero-start burn-in, 120 cycles; IV weights off
- Priors: mortality λ 1.00, structure 0.05

In 2019, modelled prevalence is 15% below GBD in Indonesia and 5% below in Uganda

	Scored				Validation, model/GBD	
	GBD prevalence	Model/ GBD	GBD deaths	Model/ GBD	All-age deaths	Incidence
Indonesia	1,296,208	0.85	5,562	1.15	1.15	1.35
Uganda	712,776	0.95	352	1.01	1.01	0.90

Layer-2 fit, 2019; targets on the GBD rate × WPP population basis; both sexes. A status of calibrated does not by itself certify the fit.

Every death age is scored, yet the model reproduces only 71% (Indonesia) and 40% (Uganda) of GBD's under-20 RHD deaths

	Indonesia	Uganda
GBD RHD deaths under 20, 2019	821	45
Share of all-age GBD RHD deaths	14.8%	12.7%
Model RHD deaths under 20	583	18
Model/GBD, deaths all ages (scored)	1.15	1.01

No death cells are masked: deaths at every age enter the loss; the age floor is read from the saved calibration.

Fitted stage-D mortality is 12.1% (Indonesia) and 6.0% (Uganda) per year, 2% above and within 1% of their provisional priors

Annual	Indonesia	Uganda
Stage-D mortality, fitted	12.10%	5.98%
Prior centre (workbook proxy)	11.90%	5.97%
Bounds	3%–25%	3%–25%
Stage-D share of living RHD, 2019	4.87%	0.91%
GBD heart-failure proxy share	3.21%	0.86%
Realised D death risk, 2019	11.94%	5.87%

One level for all ages and both sexes. Stage shares emerge from the calibrated dynamics; they are not free parameters.

A one-at-a-time sweep flags C → B (Indonesia) and C → B (Uganda) as flat

Parameter	Indonesia	Uganda
A → no RHD	0.00	0.00
A → B	0.09	0.09
B → A	0.09	0.27
B → C	0.09	0.09
C → B	**0.73**	**0.55**
C → D	0.09	0.09
Stage-D mortality	0.09	0.09
Adult progression multiplier	0.18	0.09
Adult remission multiplier	0.09	0.00

Flatness: share of a parameter's range within the near-optimal band, others held at the optimum; amber = flagged flat. Incidence multipliers in backup.

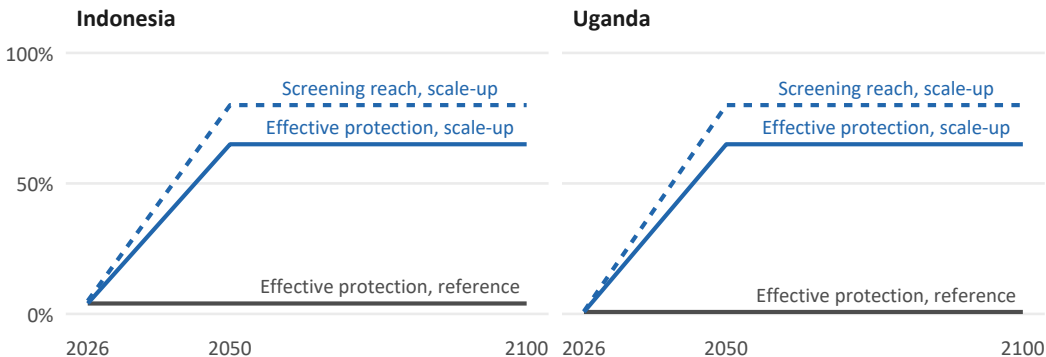
What the comparison changes

Only the care cascade differs between arms; surgery coverage is identical in both

	Indonesia		Uganda	
Cascade step (denominator)	Reference	Scale-up	Reference	Scale-up
Screening reach (ages 5–19 and 20–39)	5%	5% → 80%	1%	1% → 80%
Diagnosis (people with RHD)	5%	5% → 80%	0.9%	0.9% → 80%
SAP (all prevalent RHD)	4%	4% → 65%	0.75%	0.75% → 65%
Adherence modifier	1.00	1.00	1.00	1.00
Surgery (C, D in need)	5%	5%	3%	3%

Scale-up ramps linearly from 2026 to 2050, then holds to 2100. Surgery need per year: 3% of C and 20% of D (Indonesia); 10% and 40% (Uganda).

Treatment coverage reaches 65% by 2050 and then holds; screening never constrains it



Effective protection = min(reach, diagnosis, SAP coverage) × adherence, a simplifying reach ceiling, not a measured cascade; here it equals nominal treatment coverage (adherence 1).

At target coverage SAP cuts the A→B, B→C and C→D hazards by 62%, 52% and 36%

$$h_{ij}(c_t) = h_{ij}(0)\Big[1 - c_t\,(1 - \mathrm{HR}_{ij})\Big], \qquad c_t = c_t^{\mathrm{tx}}\,a_t, \qquad c_t^{\mathrm{tx}} = \min\left(c_t^{\mathrm{scr}}, c_t^{\mathrm{dx}}, c_t^{\mathrm{trt}}\right)$$

h_{ij}: annual hazard of progression $i \to j$. HR = 0.039 (A→B), 0.200 (B→C), 0.450 (C→D). a_t: adherence modifier. A population-average hazard approximation; no direct effect on mortality or incidence.

Costs are unit prices times modelled volumes; screening is priced per scheduled encounter at ages 5–19 and 20–39

Service	Annual volume	Indonesia	Uganda
Echo screening	Reach $\times \left[\frac{2}{15} P_{5\text{–}19} + \frac{1}{20} P_{20\text{–}39}\right]$	US$1.10	US$0.40
Diagnosis	Coverage × prevalent RHD	not costed	not costed
SAP (optimal treatment)	Coverage × all prevalent RHD	US$110	US$40
Valve surgery	Need × coverage × C, D	US$9,000	US$5,000

Unit costs per screening encounter, per person-year on SAP and per surgery; sources and price year are not recorded in the pipeline. $P_{a\text{–}b}$: population aged a–b, both sexes. Implied perspective: programme delivery cost.

Each death averted is valued at a VSL anchored on GNI per capita and grown with GDP per capita

- Benefit$_t$ = deaths averted$_t$ $\times$ VSL$_t$; VSL$_t$ = 30 $\times$ GNI pc$_{2019}$ $\times$ GDP pc$_t$ / GDP pc$_{2019}$
- GNI per capita, World Bank Atlas method, 2019: US$4,020 (Indonesia), US$810 (Uganda); VSL US$120,600 and US$24,300, growing 3.0% and 3.2% a year
- No separate GDP benefit: GDP per capita only carries VSL forward
- Coates et al. (2021) used a VSL of roughly 30 times income

Provisional multiple

The 30$\times$ multiple is carried over from the former GDP-based specification and has not been re-estimated for GNI.

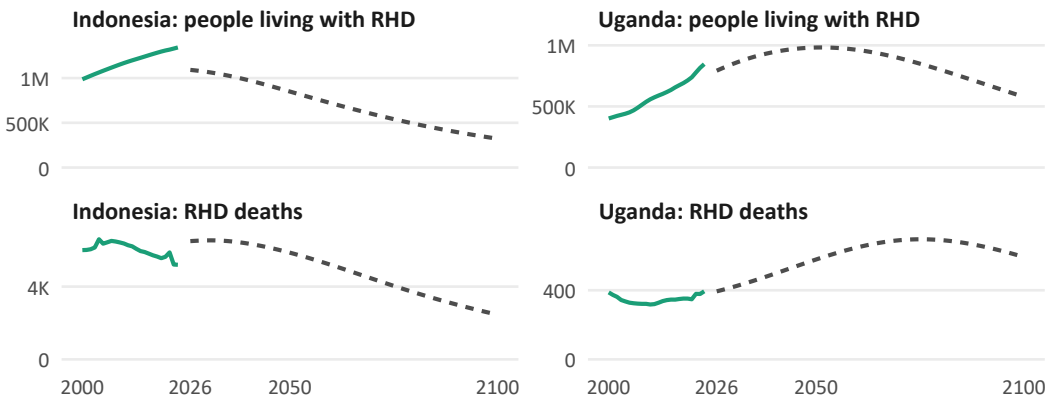
The decision metric is the discounted benefit–cost ratio over 2026–2100

$$\text{BCR} = \frac{\sum_t \delta_t \, \Delta D_t \, \text{VSL}_t}{\sum_t \delta_t \, \Delta C_t}, \qquad \text{VSL}_t = m \, \text{GNIpc}_b \, (1+g)^{t-b}, \qquad \delta_t = (1+r)^{-(t-2026)}$$

ΔD_t: RHD deaths averted (reference minus scale-up, signed). ΔC_t: incremental cost in US$; the ratio is defined only when its sum is positive, otherwise reported as cost-neutral or cost-saving. m = 30; b = 2019; r = 3% for costs and benefits. Cost per death and per disability-adjusted life year (DALY) are secondary outputs (backup).

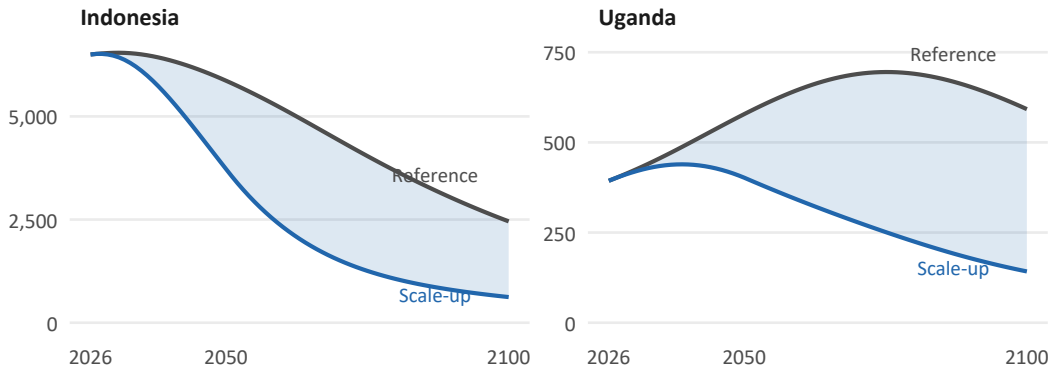
What the model projects

Projected 2026 RHD deaths are 25% above GBD 2023 in Indonesia and within 1% of in Uganda



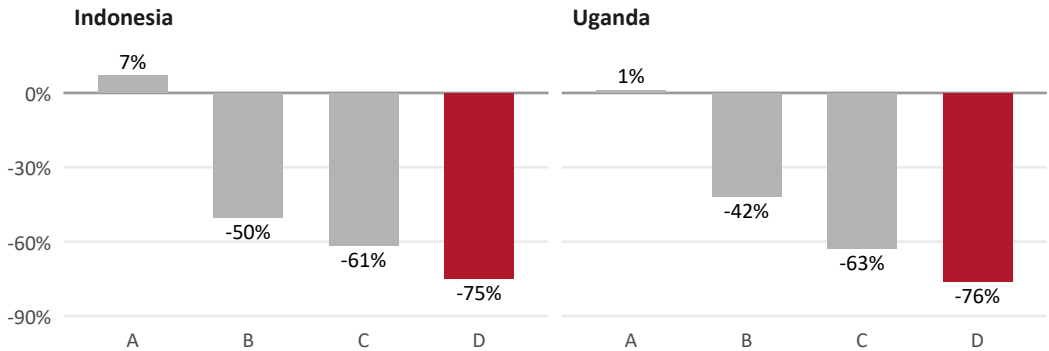
Solid green: GBD 2023 (GBD rate × WPP population), 2000–2023. Dashed grey: model reference arm, 2026–2100; prevalence on the calibration's observation convention. All ages, both sexes.

Scale-up averts about 147,900 RHD deaths in Indonesia and 21,500 in Uganda by 2100



Annual RHD deaths; shaded = difference between arms. Cumulative: 41% (Indonesia) and 48% (Uganda) of reference-arm RHD deaths, 2026–2100.

By 2100, scale-up leaves stage D 75% smaller in Indonesia and 76% smaller in Uganda



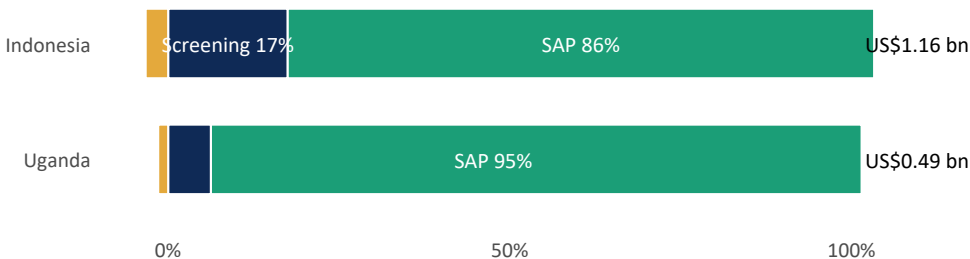
Prevalent cases by stage, scale-up vs reference, end of 2100. Total prevalent RHD: 7.0% lower (Indonesia), 3.1% lower (Uganda).

By 2050 the scale-up delivers 10.2 million screening encounters a year in Indonesia and keeps 542,800 on SAP

	Indonesia		Uganda	
Per year, 2050	Reference	Scale-up	Reference	Scale-up
Screening encounters	0.64 m	**10.19 m**	0.05 m	**3.87 m**
Prevalent RHD diagnosed	42,800	**668,100**	8,800	**781,500**
People on SAP (all ages)	34,200	**542,800**	7,400	**634,900**
Valve surgeries	561	**361**	161	**114**

Counts from the model's service trace; m = million. Encounters = reach × scheduled screens per person-year × population aged 5–19 and 20–39 (expected volumes, not unique people). Diagnosis and SAP volumes are shares of the prevalent RHD stock at all ages.

Echo screening is 17% of Indonesia's and 6% of Uganda's discounted incremental cost



Discounted (3%, base 2026) incremental cost, 2026–2100, by component; surgery is -3.3% (Indonesia) and -1.5% (Uganda).

Benefit–cost ratios are 18.85 in Indonesia and 1.46 in Uganda over 2026–2100

Discounted to 2026 at 3%	Indonesia	Uganda
Incremental cost	US$1.16 bn	US$0.49 bn
Monetised benefit	US$21.94 bn	US$0.72 bn
Net benefit	US$20.78 bn	US$0.22 bn
Benefit–cost ratio	**18.85**	**1.46**
RHD deaths averted (undiscounted)	147,911	21,514

US$; benefit = deaths averted × VSL only (30× GNI per capita, 2019, grown with GDP per capita); unit-cost price year undocumented. Deterministic point estimates.

Pricing screening per scheduled encounter raises both ratios; the GNI-based, VSL-only valuation then lowers them

	Indonesia			Uganda		
Discounted, US$ m	Cost	Benefit	BCR	Cost	Benefit	BCR
Retired definitions	6,292	23,403	3.72	1,067	1,228	1.15
Screening correction only	1,164	23,403	20.11	493	1,228	2.49
Valuation correction only	6,292	21,939	3.49	1,067	718	0.67
Both (saved run)	**1,164**	**21,939**	**18.85**	**493**	**718**	**1.46**

Retired definitions recomputed from the saved run: screening = reach × whole population every year; benefit = deaths averted × (30 + 1) × GDP per capita. Deaths averted are identical in every row; the first row reproduces the pre-revision results.

Ending the horizon in 2050, as the abstract did, lowers the ratios to 5.40 and 0.23

	Indonesia		Uganda	
	2026–2050	2026–2100	2026–2050	2026–2100
RHD deaths averted	20,900	147,900	1,500	21,500
Benefit–cost ratio	**5.40**	**18.85**	**0.23**	**1.46**
Net benefit (discounted)	US$2.52 bn	US$20.78 bn	US$-0.15 bn	US$0.22 bn

2026–2050 columns are deck-only sums of the saved runs to 2050; the model runs forward only, so this equals a 2050 run with the same inputs.

What must be resolved

The engine and pipeline are implemented; the parameters that drive deaths and costs are provisional

Implemented

- A–D engine, hazard exits
- Joint two-layer calibration
- Stage-D-only mortality
- SAP on progression; costing, BCR

Provisional

- Stage-D mortality proxy
- SAP stage mapping, C→D effect
- Childhood RHD deaths
- Screening schedule, costs, VSL multiple

Planned

- Sensitivity analyses
- Age-specific D mortality
- Adherence evidence
- Surgery pathway

Stage-D mortality depends on its proxy: the two definitions differ up to 19-fold

			Worksheet stage-D death rate	
	Fitted	Prior centre	Literature proxy	GBD heart-failure proxy
Indonesia	12.1%	11.9%	2.4%	11.9%
Uganda	6.0%	6.0%	0.3%	6.0%

Worksheet rates divide GBD 2023 RHD deaths by an estimated stage-D count (a literature stage-D share, or GBD RHD heart-failure prevalence); collaborator worksheet, October 2026. All ages, both sexes.

4 of Indonesia's and 2 of Uganda's child-age transitions fall outside published ranges

Annual probability	Literature range (studies)	Indonesia	Uganda
A → no RHD	0.007–0.249 (11)	0.093	0.122
A → B	0.002–0.072 (15)	0.063	0.004
B → A	0.004–0.203 (11)	0.173	0.012
B → C	0.001–0.125 (9)	**0.200**	0.081
C → B	0.012–0.081 (4)	**0.005**	**0.004**
C → D	0.006–0.250 (4)	**0.300**	0.112
D → C	0.004–0.071 (3)	**0.000**	**0.000**

Annualised ranges from the collaborators' September 2026 literature table; amber bold = outside the range. Values apply below age 20; adult hazards are scaled (backup).

SAP now acts on progression, but the trial evidence covers children in early stages and the C-to-D effect is assumed

In the model

- Hazard ratios 0.04 (A→B), 0.20 (B→C), 0.45 (C→D)
- No direct effect on mortality or incidence

In the sources

- Coates et al. (2021): 55% (7–78%) on progression to severe RHD
- WHF 2023: in a trial in children with stage A or B, SAP slowed progression; number needed to treat 13 (Rwebembera et al., 2023)

Mapping trial categories to WHF stages A and B is provisional; no pathway-specific trial informs C to D.

The strongest objection: totals do not identify stages, and stages drive the benefit

A skeptic would say

GBD gives only total prevalence and deaths. Many stage splits fit about equally well, yet the SAP benefit depends on how many people progress toward stage D.

What the evidence supports

Results are conditional on one calibrated stage structure. Stage-level echo targets would pin the split; until then, ranges should come from sensitivity analysis.

Remaining limitations bias results in directions we can name

Limitation	Likely effect on results	What resolves it
Childhood deaths	Child deaths averted missing	Acute childhood pathway; review
Transferability	Direction unknown	Country cohort or registry data
Adherence	Effect per dollar high	Adherence and retention data
Screening schedule	Ages, frequency assumed	Programme ages, interval data
SAP eligibility	Volume approximate	Eligibility, duration, retention
Valuation	30× GNI provisional	VSL and discount sensitivity
Uncertainty	Spread unknown	Probabilistic, one-way analyses

Five updates, in this order, should precede any investment conclusion

1. **Stage-D mortality and the childhood pathway** — deaths, stage mix, every result
2. **SAP stage mapping and the C→D effect** — size and timing of benefit
3. **Echo stage targets** — the stage split the benefit depends on
4. **Screening ages, frequencies and unit-cost sources** — screening is 6%–17% of incremental cost
5. **Sensitivity analyses** — adherence, weights, priors and the death-age floor

On current saved runs, scale-up returns US$18.9 (Indonesia) and US$1.5 (Uganda) per dollar through 2100—but stage-D mortality, the SAP effects and the screening schedule must be settled first.

Five collaborator decisions will define the next model version

1. Is stage-D-only RHD mortality acceptable, and which proxy should anchor its level?
2. How should childhood RHD deaths be represented: an acute pathway or an age-specific risk?
3. Do the GOAL trial effects map to WHF stages A and B, and what supports a C→D effect?
4. Are the assumed screening windows and frequencies (2 in 5–19, 1 in 20–39) right, and should adults be screened at all?
5. Should adherence below one be modelled, and from which data?

Backup slides hold detail for questions and sit outside the 40-minute budget

- Cycle order and full engine equations
- Calibration losses, settings, targets and fit
- Parameters, incidence multipliers, initialisation
- Near-optimal sets; literature transitions
- Economic parameters and detailed outputs
- Budget impact; abstract; discrepancies; provenance

Within each annual cycle, the engine runs seven steps in a fixed order

1. Stocks A–D enter year t already aged; $S_t = A_t + B_t + C_t + D_t$
2. Susceptible pool $W_t = \max(P_t - S_t, 0)$; incidence competes with other-cause death
3. Coverage and adherence set protection c_t; surgery reach $f_C c^{\text{surg}}$, $f_D c^{\text{surg}}$
4. Each state splits its exits on the hazard scale
5. Year-end stocks collect stays and inflows; new cases join A
6. Record deaths and volumes: screening encounters $= c_t^{\text{scr}} \sum_g (n_g / W_g)\, P_{g,t}$; diagnosed, on SAP $= c\, S_t$; surgeries $= f\, c^{\text{surg}} \times$ C, D
7. Survivors age one year; age 0 starts empty; 100+ is open

The four stage updates share one form; only stage D lacks an exit to a higher stage

$$
\begin{aligned}
A_t^{+} &= A_t\, e^{-H_A} + B_t\, p_{B\to A} + W_t\, p_{W\to A} \\
B_t^{+} &= B_t\, e^{-H_B} + A_t\, p_{A\to B} + C_t\, p_{C\to B} \\
C_t^{+} &= C_t\, e^{-H_C} + B_t\, p_{B\to C} + D_t\, p_{D\to C} \\
D_t^{+} &= D_t\, e^{-H_D} + C_t\, p_{C\to D}
\end{aligned}
$$

Each p is the hazard-split outflow. Before the split, p_{CD} is scaled by $1 - \rho_{CD}^{\text{surg}} f_C c^{\text{surg}}$ and the A→B, B→C, C→D hazards by $1 - c_t(1 - \text{HR})$; m_D by $1 - \rho_{Dd}^{\text{surg}} f_D c^{\text{surg}}$. RHD deaths $= \sum_s s_t\, p_{s\to\text{RHD}}$. $A_{t+1} = \text{shift}(A_t^{+})$.

Layer 1 minimises a weighted log error with priors and smoothness on band multipliers

$$L_1 = \sum_{X \in \{I,P,D\}} w_X \frac{\sum_j \omega_j \left[\ln(X_j^{\text{mod}} + \varepsilon) - \ln(X_j^{\text{GBD}} + \varepsilon) \right]^2}{\sum_j \omega_j} + \lambda \sum_g \left[\frac{\eta_g^2}{\sigma_\alpha^2} + \frac{\zeta_g^2}{\sigma_\beta^2} \right] + \lambda_s \sum_g \left[(\Delta^2 \eta_g)^2 + (\Delta^2 \zeta_g)^2 \right]$$

$\eta_g = \ln \alpha_g$ (incidence), $\zeta_g = \ln \beta_g$ (mortality), bands 0-14, 15-24, 25-44, 45-64, 65+. Weights w = 1, 1, 2; ω inverse-variance from GBD intervals (used). σ_α = 0.75, σ_β = 0.5. multistart L-BFGS-B, 8 random starts plus baseline. Only α is used, as Layer-2 start values.

Layer 2 minimises weighted log errors on masked cells plus mortality and structural priors

$$L_2 = w_P \bar{\ell}_P + w_D \bar{\ell}_D + w_I \bar{\ell}_I + w_S \bar{\ell}_S + \lambda_m \frac{(\ln m_D - \ln m_D^0)^2}{\sigma_m^2}$$
$$+ \lambda_s \Big[\sum_k \frac{(\text{logit}\, p_k - \text{logit}\, p_k^0)^2}{\sigma_p^2} + \sum_{b,s} \frac{(\ln \alpha_{bs})^2}{\sigma_\alpha^2} + R_{\text{age}} \Big] + \lambda_\Delta \sum_{s,b} (\Delta^2 \ln \alpha_{bs})^2$$

$\bar{\ell}_X$: weighted mean of $[\ln(X^{\text{mod}} + \varepsilon) - \ln(X^{\text{tgt}} + \varepsilon)]^2$ over unmasked cells, both sexes pooled. w_P, w_D, w_I, w_S = 2.5, 1.0, 0.0, 1.0; λ_m = 1.00, λ_s = 0.05, λ_Δ = 0.5; σ_m = 0.35, σ_p = 2, σ_α = 1. Settings as saved for Indonesia; Uganda's are identical (checked in the next slide).

Scored cells are all-age deaths and all-age prevalence in 2010–2019; the rest are validation

	Indonesia			Uganda		
Cell status	Prev.	Deaths	Incid.	Prev.	Deaths	Incid.
Below count threshold	20	5	0	20	30	0
Held-out year	440	440	440	440	440	440
Missing (masked)	0	20	0	0	20	0
Scored	420	415	0	420	390	0
Validation only	0	0	440	0	0	440

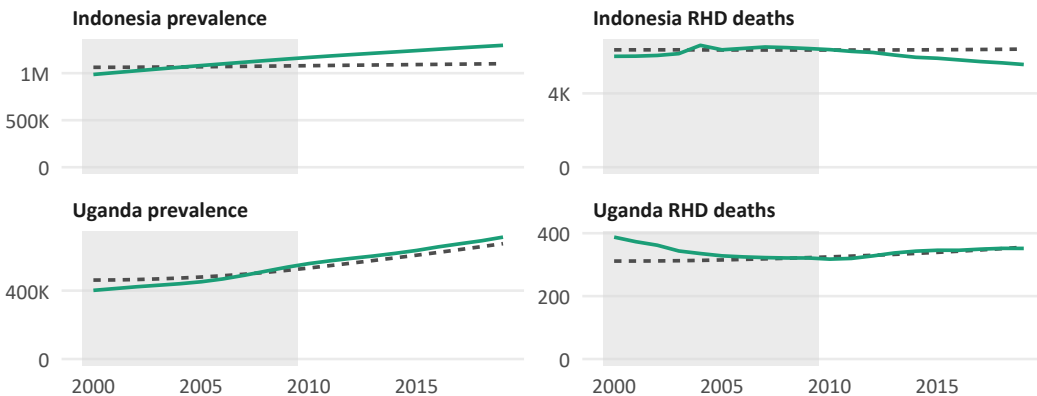
Age-group × sex × year cells, 2000–2019. Objective settings are identical in the two countries. Masking as recorded: missing GBD cells masked (NA, never zero-filled); counts < 1 masked; years < 2010 weight 0 (held-out); eps = 1; death-target age floor = 0 (0 cells masked, reported as validation).

Layer-2 targets use GBD rates times WPP population, which differ from published counts by up to 10%

	Indonesia		Uganda	
Measure	Range	Window	Range	Window
Deaths	0.958–1.020	0.973	0.977–1.084	1.031
Incidence	0.964–1.033	0.997	0.933–1.013	0.981
Prevalence	0.966–1.025	0.983	0.904–1.014	0.974

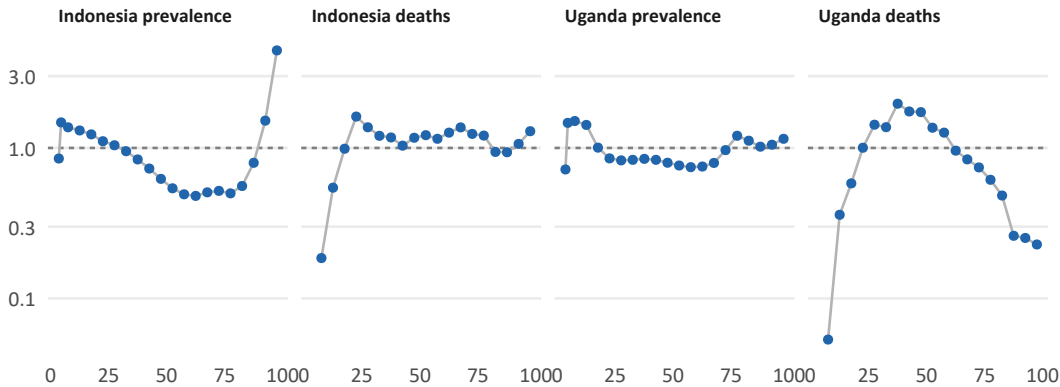
Ratio of GBD published Number to GBD Rate × WPP population, by sex and year, 2000–2019. Selected basis: GBD rate × WPP population; model/target ratios elsewhere use that basis.

Prevalence and deaths are fitted in 2010–2019; earlier years are projected without weight



Green: GBD target (GBD rate × WPP population). Dashed grey: Layer-2 model. Grey band: held-out years. Prevalence observed on the mid convention.

By age, the fit leaves childhood deaths and some adult prevalence ages far from GBD



Model / GBD target by age group, 2019, log scale. Hollow amber: not scored. Age on the horizontal axis (group midpoint).

Calibrated transitions, stage-D mortality and adult multipliers, with priors and bounds

	Indonesia		Uganda	
Annual probability or multiplier	Prior [bounds]	Fitted	Prior [bounds]	Fitted
A → no RHD	0.0500 [0.005, 0.600]	0.0931	0.0500 [0.005, 0.600]	0.1222
A → B	0.0200 [0.001, 0.250]	0.0634	0.0200 [0.001, 0.250]	0.0036
B → A	0.0100 [0.002, 0.400]	0.1727	0.0100 [0.002, 0.400]	0.0123
B → C	0.0300 [0.001, 0.200]	**0.2000**	0.0300 [0.001, 0.200]	0.0805
C → B	0.0050 [0.001, 0.150]	0.0051	0.0050 [0.001, 0.150]	0.0043
C → D	0.0600 [0.001, 0.300]	**0.3000**	0.0600 [0.001, 0.300]	0.1120
D → C	fixed	0.0000	fixed	0.0000
Stage-D mortality	0.1190 [0.030, 0.250]	0.1210	0.0597 [0.030, 0.250]	0.0598
Adult progression multiplier	1.0000 [0.050, 3.000]	0.3381	1.0000 [0.050, 3.000]	0.5108
Adult remission multiplier	1.0000 [0.050, 3.000]	0.8518	1.0000 [0.050, 3.000]	0.6969

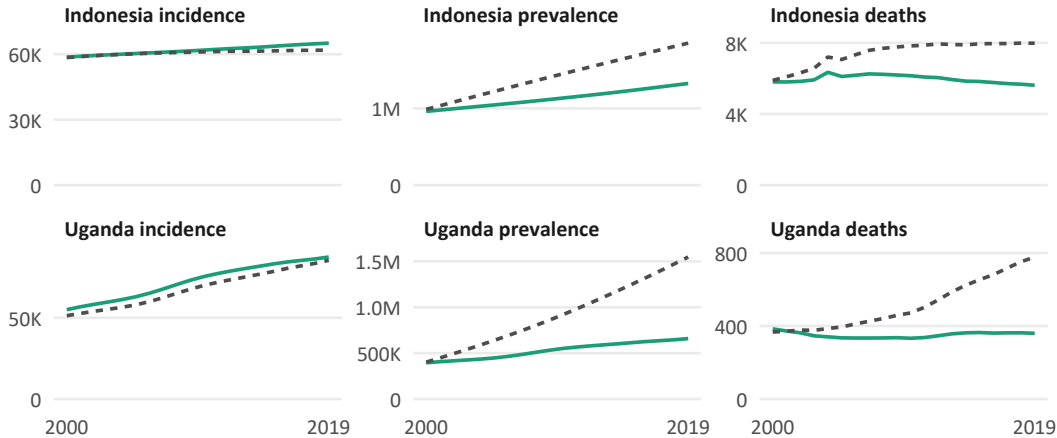
Transitions apply below age 20; multipliers act on adult hazards, $p_{\text{adult}} = 1 - (1 - p)^{\kappa}$. Amber bold: at a bound. Stage A–C RHD mortality is exactly zero; residual multiplier fixed at 1.

Layer-2 incidence multipliers move away from the Layer-1 start values

	Indonesia		Uganda	
Age band	Layer-1 start	Layer 2 (used)	Layer-1 start	Layer 2 (used)
0-14	0.98 / 0.97	1.36 / 1.14	0.92 / 0.91	0.83 / 0.72
15-24	0.98 / 0.98	1.43 / 1.28	0.95 / 0.94	1.00 / 0.91
25-44	0.99 / 0.98	1.45 / 1.39	0.96 / 0.96	1.13 / 1.09
45-64	1.00 / 0.99	1.36 / 1.38	0.98 / 0.98	1.21 / 1.22
65+	1.00 / 1.00	1.25 / 1.33	1.00 / 1.00	1.26 / 1.32

Female / male multipliers on the GBD incidence pattern. Only the Layer-2 values enter the projection, applied once inside the saved 2019 per-susceptible incidence.

Layer 1 is a diagnostic proxy; its fit does not feed the projection



Green: GBD published counts. Dashed grey: Layer-1 aggregate proxy (no remission). Summed over age groups and sex.

The burn-in converges and the bridge carries the calibrated state to 2026

	Indonesia	Uganda
Burn-in cycles / doubled check	120 / 240	120 / 240
Largest stage change, doubled burn-in	4.3e-10	3.2e-09
Bridge prevalence / GBD, 2023	0.82	0.88
Bridge RHD deaths / GBD, 2023	1.24	0.96
Seed entering 2026 / legacy GBD-rate seed	0.79	0.84

Zero-start burn-in under 2000 inputs; bridge 2020–2025 under reference care. GBD comparisons use GBD rate × WPP population. The seed is not reset to GBD prevalence.

The near-optimal solutions (5 Indonesia, 5 Uganda) are alternative optima, not a posterior

Country	Objective	m_D	A→no RHD	B→C	C→D
Indonesia	1.8653	0.1210	0.0931	0.2000	0.3000
Indonesia	1.8653	0.1210	0.0931	0.2000	0.3000
Indonesia	1.8653	0.1210	0.0925	0.2000	0.3000
Indonesia	1.8653	0.1210	0.0926	0.2000	0.3000
Indonesia	1.8653	0.1210	0.0931	0.2000	0.3000
Uganda	0.7020	0.0598	0.1217	0.0807	0.1117
Uganda	0.7020	0.0598	0.1224	0.0808	0.1117
Uganda	0.7020	0.0598	0.1226	0.0808	0.1114
Uganda	0.7020	0.0598	0.1222	0.0805	0.1120
Uganda	0.7020	0.0598	0.1217	0.0812	0.1115

Prior-centre objective: 8.74 (Indonesia), 10.98 (Uganda); minimum 1.865 and 0.702. The near-optimal band width is not recorded in the saved bundle.

Published annual transition probabilities vary 7- to 108-fold across studies

Transition	Studies	Minimum	Maximum	Max / min	Indonesia	Uganda
A → no RHD	11	0.0074	0.2490	34	0.0931	0.1222
A → B	15	0.0015	0.0723	48	0.0634	0.0036
B → A	11	0.0042	0.2025	48	0.1727	0.0123
B → C	9	0.0012	0.1253	108	0.2000	0.0805
C → B	4	0.0119	0.0810	7	0.0051	0.0043
C → D	4	0.0057	0.2497	44	0.3000	0.1120
D → C	3	0.0042	0.0710	17	0.0000	0.0000

Table 2 of the collaborators' September 2026 worksheet: each study's probability annualised as $1 - (1 - p)^{1/\text{years}}$. Includes both arms of one SAP trial (control and perfect adherence).

Economic parameters differ by country in GNI, growth and unit costs

Parameter	Indonesia	Uganda
Discount rate	3%	3%
Discount base year	2026	2026
GNI per capita, Atlas method (VSL anchor)	US$4,020	US$810
GNI / VSL anchor year	2019	2019
VSL multiple of GNI per capita	30	30
VSL in the anchor year	US$120,600	US$24,300
Real GDP per capita growth (carries VSL)	3.0%	3.2%
DALYs per RHD death	30	30
Cost per screening encounter	US$1.10	US$0.40
Cost per person-year on SAP	US$110	US$40
Cost per surgery	US$9,000	US$5,000

As saved with the economic results. GNI: World Bank WDI NY.GNP.PCAP.CD, current US$ of the anchor year. DALY: disability-adjusted life year (deaths only, no years lived with disability).

Cost-effectiveness outputs exist but mix discounted and undiscounted terms

2026–2100	Indonesia	Uganda
Cost per death averted	US$7,867	US$22,935
Cost per DALY averted	US$851	US$2,994
VSL in 2026 (saved)	US$148,323	US$30,295
Mean annual incremental cost per person (deck-only)	US$0.13	US$0.19

Cost per death: discounted cost / undiscounted deaths averted. Cost per DALY: discounted cost / (discounted deaths averted × 30). Per-person cost: undiscounted incremental cost / population, averaged over years.

Annual incremental cost peaks in 2050 (Indonesia) and 2050 (Uganda); the scale-up ramp ends in 2050

	Indonesia, US$ m			Uganda, US$ m		
Year	Reference	Scale-up	Increment	Reference	Scale-up	Increment
2030	11.1 m	24.9 m	13.8 m	0.9 m	4.7 m	3.8 m
2040	10.5 m	54.2 m	43.6 m	1.0 m	15.9 m	14.9 m
2050	9.5 m	74.2 m	64.7 m	1.1 m	27.5 m	26.4 m
2060	8.3 m	62.1 m	53.8 m	1.2 m	26.9 m	25.7 m
2070	7.1 m	51.9 m	44.8 m	1.2 m	25.0 m	23.8 m
2080	5.9 m	43.4 m	37.5 m	1.2 m	22.3 m	21.1 m
2090	4.9 m	36.3 m	31.4 m	1.1 m	19.3 m	18.2 m
2100	4.1 m	30.5 m	26.4 m	1.0 m	16.3 m	15.3 m

Undiscounted annual programme cost by arm (screening + SAP + surgery), selected years.

The submitted abstract predates the current specification; truncated saved runs no longer reproduce it

	Uganda (low-income illustration)		Indonesia (upper-middle-income)	
	Abstract	Saved run	Abstract	Saved run
RHD deaths averted by 2050	7,900	1,500	23,000	20,900
Death-rate reduction in 2050	30%	31%	24%	37%
Benefit–cost ratio	1.28	0.23	1.42	5.40

Abstract values parsed from the submitted WCRHD 2026 abstract, which presented hypothetical countries from earlier runs. Saved-run values are sums to 2050 of the current saved outputs.

No sensitivity analysis exists yet; these analyses are planned

- **Probabilistic:** sample transitions, stage-D mortality and incidence multipliers within prior ranges
- **One-way:** SAP hazard ratios and C→D effect; adherence below one; VSL multiple and GNI anchor year; discount rate; screening ages and frequencies; unit costs
- **Calibration:** prevalence : death weight ratio; structural prior weight; death-target age floor; target basis
- **Structural:** all-stage mortality comparator; direct-mortality SAP comparator; surgery returning patients to C

The pipeline is deterministic; none of the results in this deck carry uncertainty intervals.

Seven discrepancies between documents, configuration and code are logged

Item	Detail	Action
Indonesia incidence trend	Saved run 0.985; scripts/00_run_all.R 0.985 (consistent)	None
Uganda GDP per capita	Growth-index level US$1,340; World Bank 2019 is about US$822. The level cancels in VSL, which now uses GNI	Confirm source
State diagram	docs/ PNG shows SAP on stage mortality and other-cause death from A only; deck draws its own	Redraw
Layer-1 role in comments	00 block F says Layer-1 CF anchors Layer 2; 04b uses Layer-1 α only as starts	Update comments
Seed stage split comment	03 still documents a Cannon split; 05 seeds the 04b bridged state	Update comments
Diagnosis cost	Volumes tracked in 06; no unit cost in 08	Decide
Screening check in tests	tests/verify_abcd_pipeline.R check 12 still expects whole-population screening	Update test

Provenance (1/2): calibration, validation and health results trace to saved objects

Result	Script	Artifact	Object	Shape	Transform
Fit, statuses	04b	data/<C>/calibrated_rhd_parameters.rds	$stage_calibration$fit_by_group_year, $statuses	cells by age group, sex, year	sums; ratios
Parameters, priors	04b	same	$stage_calibration$parameters, $priors, $bounds	19 rows	back-transformed bounds
Identifiability	04b	same	$stage_calibration$identifiability	19 rows	none
Validation	01, 02, 06	data/<C>/temp_baseline_rates_gbd.rds; pop_observed.rds; output/<C>/out_model/<C>.rds	GBD Rate; Nx; $wsd, $stages (ref)	age × sex × year	rate × population; sums
Deaths averted	06	output/<C>/out_model/<C>.rds	wsddead	101 ages × 2 sexes × 75 years × 2 arms	ref — sap; cumulative
Stage stocks, flows	06	same	$stages A–D, flows	same; persons	sums; percent difference

Provenance (2/2): service, cost and economic results trace to saved objects

Result	Script	Artifact	Object	Shape	Transform
Coverage, seed, scenarios	05	data/<C>/baseline_state.rds	$states[[1]]$coverage, $seed	vectors of 75 years	none
Service volumes	06	output/<C>/out_model/<C>.rds	$stages n_screened, n_diagnosed, n_on_optimal_treatment, total_surgeries; $diag screening_by_year	persons or surgeries per year	sum by arm and year
Cost composition	06, 08	same + rhd_economic_results.rds	$stages volumes × $params unit costs	US$	× discount factor; sap — ref
Economic summary	08	output/<C>/tables/rhd_economic_results.rds	$summary	1 row per location	none
Horizon	08	same	$incremental	75 years	sums to 2050
Literature, stage-D proxies	none	dev/*.xlsx	Table 2; All-ages rows	probabilities	min, max; read as saved
Abstract values	none	dev/Abstract WCRHD26 DW_NM.docx	Results paragraph	text	regular expressions

In paths, <C> is Indonesia or Uganda. All paths relative to the repository root; the deck reads these files and writes nothing.

Rendering needs only the saved artifacts, R and a XeLaTeX toolchain

- Render: `rmarkdown::render("reports/slideck-technical.Rmd")` with XeLaTeX and Calibri; set `RHD_REPO_ROOT` when rendering a copy outside the repository
- Inputs: artifacts of scripts 01–08 for both countries; a missing file stops the render and names its producing script
- Checks: location, scenarios, years, specification, calibrated parameters, seed, reference care, SAP trace and economics agree for each country
- Software: R version 4.5.1 (2025-06-13 ucrt); rmarkdown 2.29; knitr 1.50; data.table 1.17.8; ggplot2 4.0.3; kableExtra 1.4.0
- Saved: calibration 2026-10-22 / 2026-10-22; model outputs 2026-10-22 / 2026-10-22 (Indonesia / Uganda)

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